End-to-End Life Expectancy Analysis

Data Cleaning, Exploratory Data Analysis and Visual Reporting

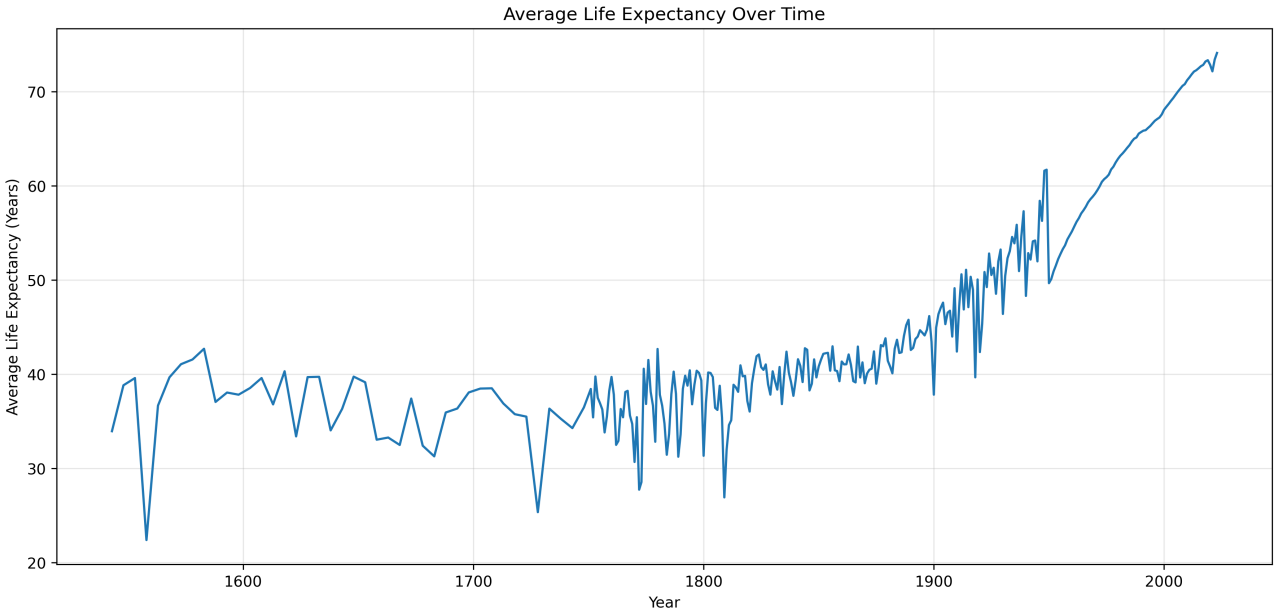
Project Overview

This project analyzes historical life expectancy data through data cleaning, exploratory analysis, statistical summaries and visualizations. The analysis covers 21,565 original records and produces a cleaned country/entity-level dataset for further analysis.

Key Performance Indicators

Metric	Value
Original records	21,565
Cleaned records	20,217
Entities	247
Years covered	1543–2023
Overall mean	62.07 years
Overall median	64.79 years
Highest average entity	Monaco — 78.05 years
Lowest average entity	South Sudan — 37.84 years
2023 average	74.12 years

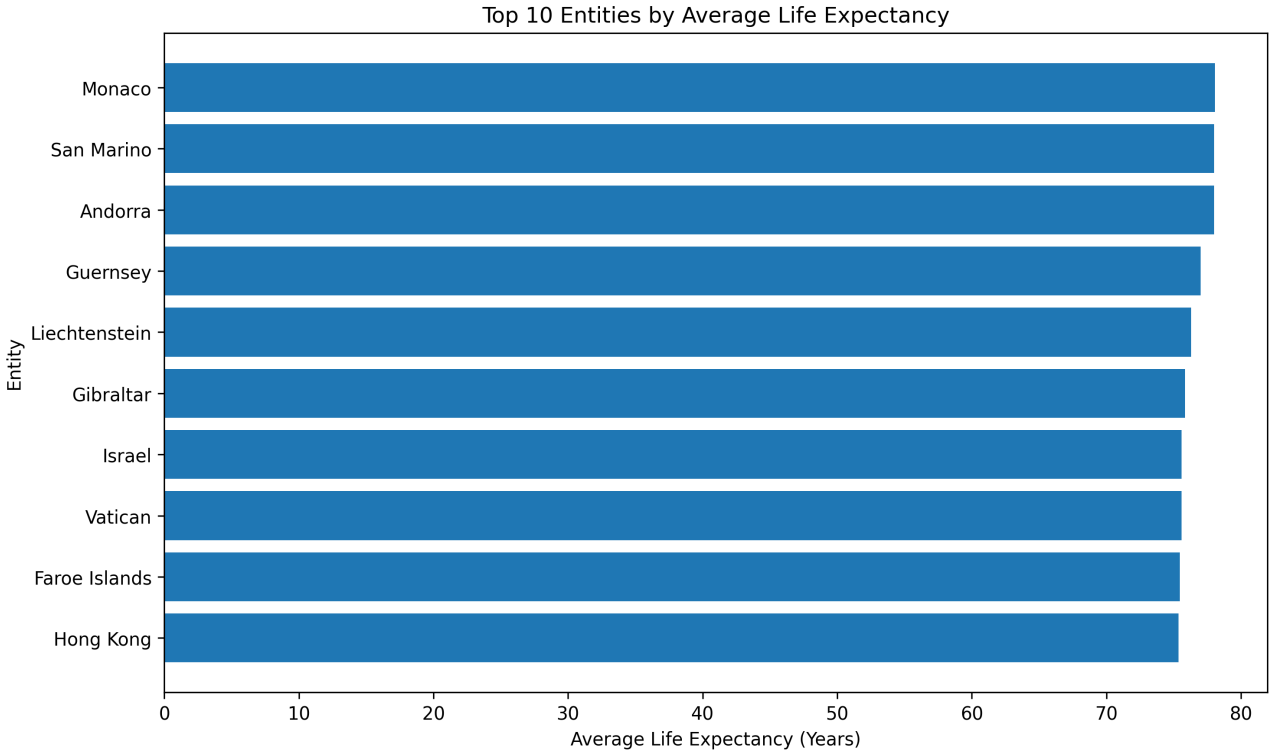
Visualization 1 — Average Life Expectancy Over Time



The later portion of the dataset shows a substantially higher average life expectancy than many historical periods. The number of entities represented also changes across time, so early historical fluctuations should be interpreted with caution.

Entity-Level Comparison

Visualization 2 — Top 10 Entities by Average Life Expectancy



Monaco has the highest average life expectancy among the entities analyzed, at 78.05 years. South Sudan has the lowest average at 37.84 years. The difference between these two averages is approximately 40.21 years.

Three Actionable Insights

1. Monitor long-term improvement: The latest yearly average is 74.12 years in 2023. Long-term monitoring can help identify periods of improvement and areas requiring additional health and social investment.

2. Address large performance gaps: The difference between the highest and lowest entity averages is approximately 40.21 years. This highlights the importance of identifying regions requiring stronger healthcare, prevention and social-development interventions.

3. Consider data coverage: The number of represented entities changes considerably across the historical period. Comparisons should therefore consider dataset coverage to avoid confusing changes in coverage with changes in overall life expectancy.

Methodology

The dataset was checked for missing values and duplicate records. Records without country/entity codes were removed for the cleaned country-level dataset. Country/entity statistics were calculated using entities with at least 20 recorded years. Two visualizations were produced to communicate the overall trend and entity-level comparison.

Conclusion: The analysis demonstrates substantial variation in life expectancy across entities and historical periods. Combining trend analysis, entity-level comparisons and data-coverage awareness provides a more reliable basis for interpreting the dataset.